SOFTWARE TESTING PRACTICAL REPORT

TUTORIAL – 13
Selenium Environment Setup

Python • Selenium WebDriver • Google Chrome • Website Navigation

Student Name	Harshita Malviya
Student ID	2504030100426
Practical	Tutorial 13 — Selenium Environment Setup
Project Name	Selenium Environment Setup
Tools	Python, PyCharm, Selenium WebDriver, Google Chrome
Target Website	https://www.selenium.dev/

Prepared using the supplied Tutorial 13 lab manual, Python script, and browser screenshots.

1. Aim

To create and configure the Python, Selenium WebDriver, PyCharm IDE, and browser environment, then verify the configuration by executing a basic Selenium program.

2. Practical Overview

The practical uses Python and Selenium WebDriver to launch Google Chrome, maximize the browser window, open the Selenium website, print the browser launch message, page title and current URL, then close the browser. The supplied lab manual describes this as a basic verification of the Selenium environment.

3. Test Environment

Parameter	Details
Programming language	Python
IDE	PyCharm
Automation library	Selenium WebDriver
Browser	Google Chrome
Target URL	https://www.selenium.dev/
Test script	tutorial13.py (as described in the lab manual)
Wait behavior in submitted code	time.sleep(60) keeps the browser open for 60 seconds before closing

4. Procedure

• Open PyCharm and select the Python project/interpreter.

• Install Selenium using `pip install selenium` in the project terminal.

• Create a Python file and import Selenium WebDriver.

• Create the Chrome WebDriver instance and maximize the browser window.

• Navigate to `https://www.selenium.dev/`.

• Print the browser launch confirmation, page title, and current URL.

• Keep the browser open for 60 seconds to allow inspection/screenshots, then close it with `driver.quit()`.

5. Test Case

Field	Details
Test Case ID	TC-13-01
Requirement ID	REQ-13-01
Feature	Browser launch and website navigation
Objective	Verify that Selenium can launch Chrome and open the target website
Preconditions	Python, PyCharm, Chrome, and Selenium are available
Test Data	https://www.selenium.dev/
Expected Result	Chrome launches and the Selenium website opens
Observed Evidence	The supplied screenshots show a browser loading state and the inventory/products page of SauceDemo, not the Selenium website
Status	Partially evidenced — see note below

Evidence note: The provided code opens selenium.dev, but the supplied browser screenshots show saucedemo.com (one loading state and one inventory page). They are included transparently as supplied browser evidence and are not presented

as proof that selenium.dev was opened by this exact script. Capture a screenshot of selenium.dev and the matching PyCharm console for complete verification.

6. Source Code Reviewed

```
from selenium import webdriver
import time

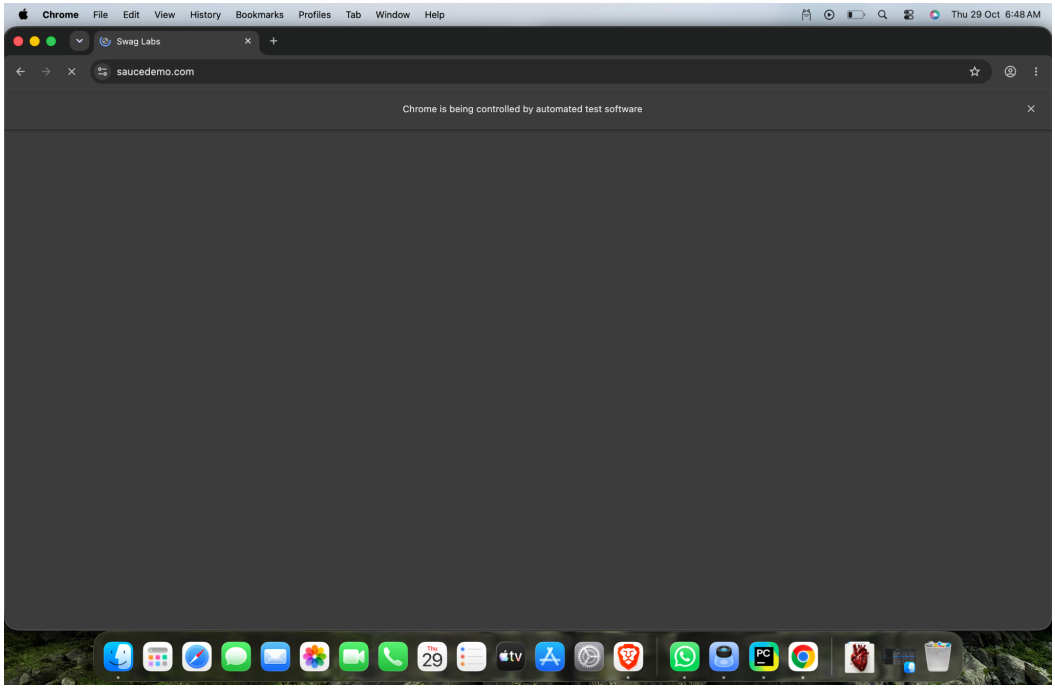
driver = webdriver.Chrome()
driver.maximize_window()
driver.get("https://www.selenium.dev/")

print("Browser launched successfully")
print("Page Title:", driver.title)
print("Current URL:", driver.current_url)

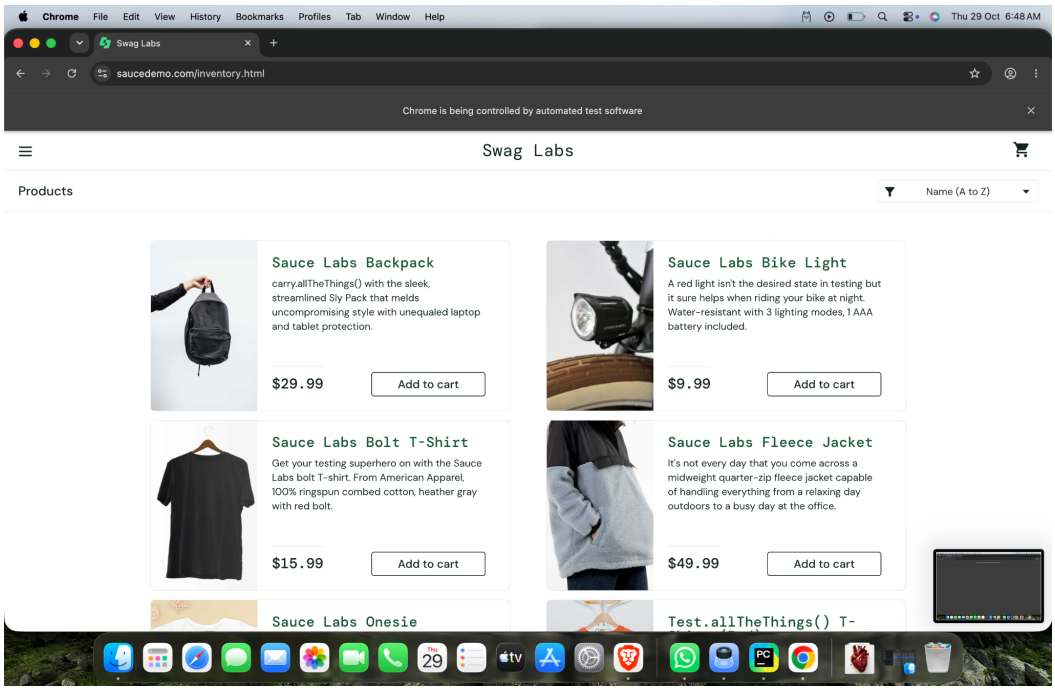
time.sleep(60)
driver.quit()
```

7. Screenshot Evidence

Screenshot 1 — Browser loading state (supplied)



Screenshot 2 — SauceDemo inventory page (supplied)


Chrome File Edit View History Bookmarks Profiles Tab Window Help
Thu 29 Oct 6:48 AM
Swag Labs
saucedemo.com/inventory.html
Chrome is being controlled by automated test software
Swag Labs
Products
Name (A to Z)
Sauce Labs Backpack
carry.allTheThings() with the sleek, streamlined Sly Pack that melds uncompromising style with unequaled laptop and tablet protection.
$29.99
Add to cart
Sauce Labs Bike Light
A red light isn't the desired state in testing but it sure helps when riding your bike at night. Water-resistant with 3 lighting modes, 1 AAA battery included.
$9.99
Add to cart
Sauce Labs Bolt T-Shirt
Get your testing superhero on with the Sauce Labs bolt T-shirt. From American Apparel, 100% ringspun combed cotton, heather gray with red bolt.
$15.99
Add to cart
Sauce Labs Fleece Jacket
It's not every day that you come across a midweight quarter-zip fleece jacket capable of handling everything from a relaxing day outdoors to a busy day at the office.
$49.99
Add to cart
Sauce Labs Onesie
Test.allTheThings() T-

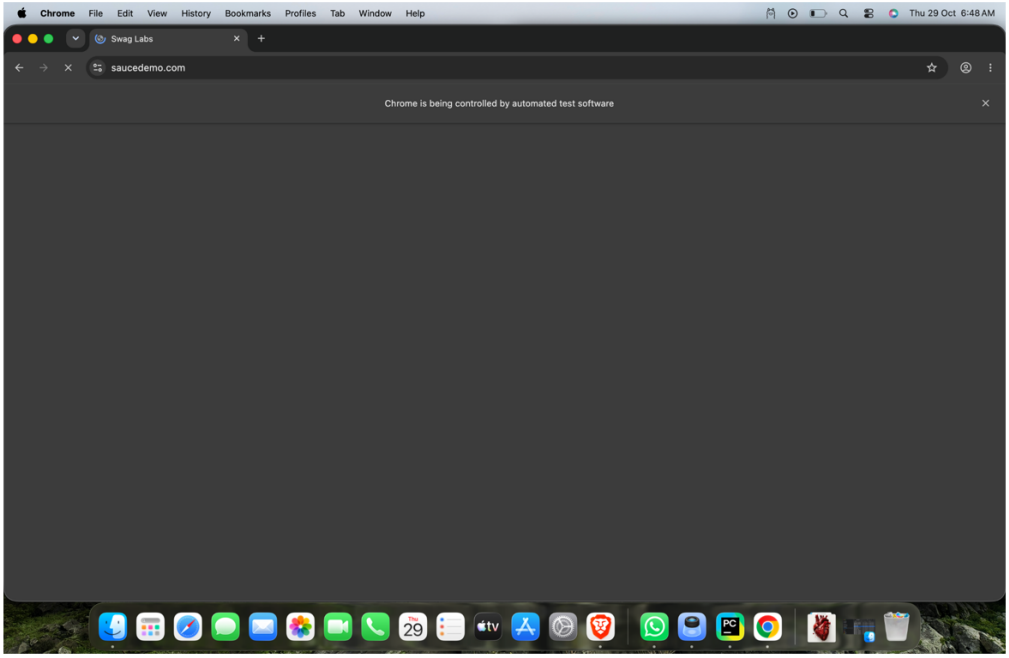
8. Screenshot Collage

A single evidence sheet combining the two screenshots supplied for Tutorial 13.

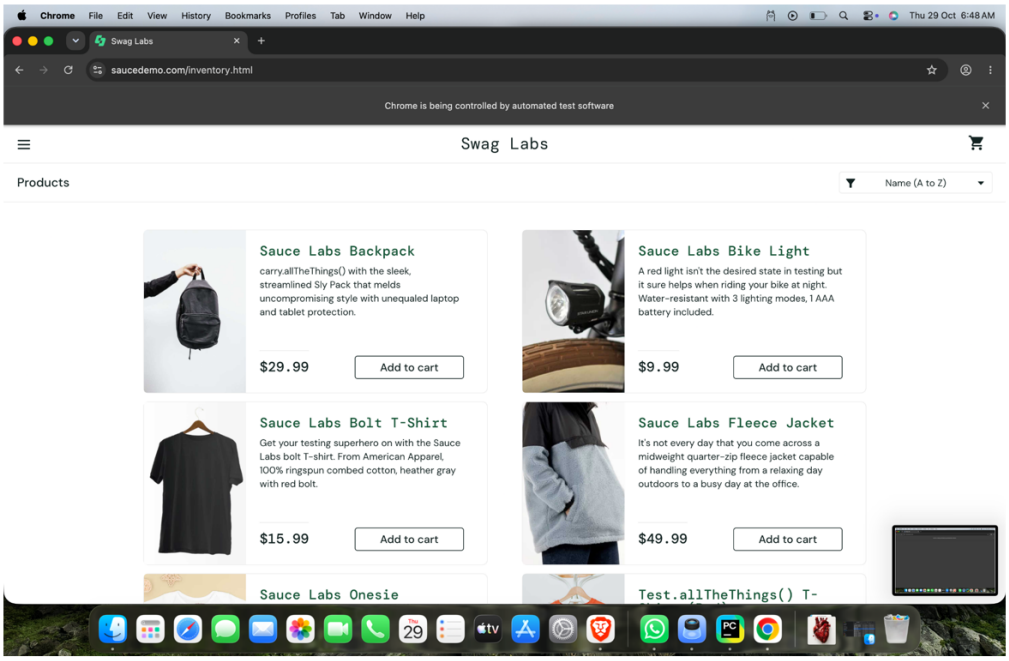
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Browser launch and Selenium website navigation evidence

Screenshot 1 ⊡ Browser loading state



Screenshot 2 ⊡ Selenium website opened



9. Test Execution and Observations

No.	Observation	Result / Note
1	Chrome automation banner is visible	Browser appears controlled by automation

No.	Observation	Result / Note
2	One screenshot shows a loading state at saucedemo.com	Loading state only; not the requested selenium.dev URL
3	Another screenshot shows saucedemo.com/inventory.html	A SauceDemo products page is visible
4	Script prints title and current URL	Console screenshot for this exact script was not supplied
5	Script closes Chrome after 60 seconds	Cleanup step is present in code

10. Result and Conclusion

The submitted program is a basic Selenium environment-check script: it launches Chrome, opens the configured URL, prints page information, waits for 60 seconds, and closes the browser. The supplied screenshots do not match the URL in the script, so they cannot independently confirm that selenium.dev was opened. A screenshot of the Selenium website and the matching console output would complete the evidence.

11. Professional QA Recommendations

• Add an assertion such as checking that the current URL contains selenium.dev.

• Capture and retain the console output showing the page title and current URL.

• Use try/finally around browser operations so the browser is closed if navigation or validation fails.

• For repeatable automated tests, consider an explicit wait for a known page element instead of relying only on time.sleep(60).